

Error Performance Degradation in Communication Systems

The task of the detector is to retrieve the bit stream from the received waveform as error free as possible, irrespective of the problems which the signal may have faced. There are 2 primary causes of error performance ~~the~~ degradation.

36

(a) Inter-Symbol Interference (ISI): to be discussed later.

(b) Electrical Noise and Interference produced by a variety of sources. These sources could be cosmic noise, atmospheric noise, switching transients, intermodulation ~~contants~~ noise, as well as interfering signals from other sources. With proper precautions much of the noise and interference entering a receiver can be reduced in intensity and even eliminated. However, there is one noise that cannot be ~~be~~ eliminated and that is the noise caused by the thermal motion of electrons in any conducting media. This motion produces thermal noise in the circuits and corrupts the signal in additive fashion.

Statistics of such thermal noise have been evaluated using quantum mechanics and it is observed that this noise is Gaussian in nature.

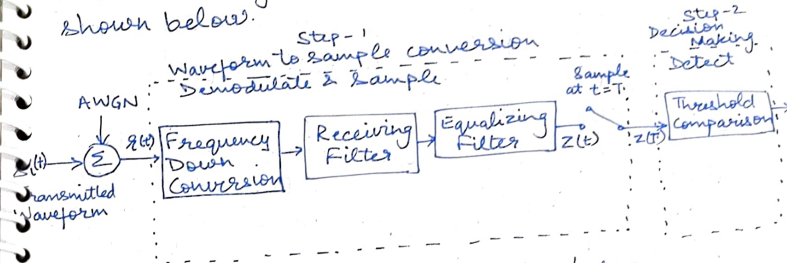
What is known about Gaussian functions is that they have a constant power spectral density in the range 0 Hz (DC) to 10^{12} Hz. Thus such a noise power ~~characteristic~~ characterized by such a constant power spectral density, may be termed as White noise.

Collectively, these characteristics of thermal noise ~~are~~ are termed AWGN (Additive White Gaussian Noise). Since thermal noise is present in almost all systems and is the predominant source of noise, AWGN modelling is the most ~~often~~ often used ~~noise~~ noise modelling in the detection processes and design of receivers.

37

Maximum Likelihood Receiver

Architecture of a maximum likelihood receiver is shown below:



The transmitted sequence is of the form

$$* \quad S_i(t) = \begin{cases} S_1(t) & 0 \leq t \leq T \text{ for a binary 1} \\ S_2(t) & 0 \leq t \leq T \text{ for a binary 0.} \end{cases}$$

Assuming an AWGN channel, i.e. a channel free from all other sources of error except the thermal noise, the received signal may be written as

$$r(t) = S_i(t) + n(t) \quad i = 1 \text{ or } 2.$$

Note: Here we are assuming a general system where the transmitted signal may be baseband modulated or bandpass modulated. Baseband system would have S_1 & S_2 as DC levels while Bandpass signalling would involve separate symbolic signals for the 2 bits.

Frequency Down conversion

This is used for band pass signals generally operating at some RF (radio frequency). In case of base band signals this block can be dispensed with. (38)

Receiving Filter

This is a compulsory part of and essentially the demodulator of the receiver, performing waveform recovery for detection.

Equalizing filter

This is used to eliminate the effect of ISI. Again if channel is said to be free from this defect, we can skip the use of this filter.

At the end of each sample point, we will have a value coming out of the demodulator and sample block of the receiver. This value is the sample $z(t)$ and is sometimes called the test statistic.

This may be expressed as

$$z(t) = a_i(t) + n_o(t) \quad i=1,2$$

where $a_i(t)$ is the desired bit or signal component and $n_o(t)$ is the noise. we may also write it simply as

$$z = a_i + n_o$$

Here a_i is expected to be a DC level and n_o is a Gaussian noise assumed to be with zero mean. Thus $z(t)$ is also a Gaussian function with mean a_i .

Thus, as the pdf of noise (Gaussian) is expressed as

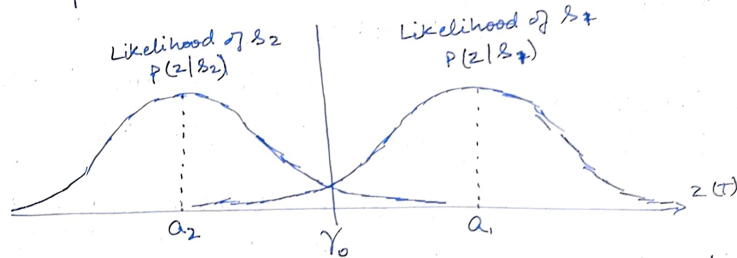
$$p(n_o) = \frac{1}{\sigma_o \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{n_o}{\sigma_o}\right)^2} \quad (39)$$

where σ_o^2 is the noise variance, the noise probability for the cases when s_1 is transmitted and when s_2 is transmitted can be written as

$$p(z|s_1) = \frac{1}{\sigma_o \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{z-a_1}{\sigma_o}\right)^2} \quad \text{and}$$

$$p(z|s_2) = \frac{1}{\sigma_o \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{z-a_2}{\sigma_o}\right)^2}$$

Here $p(z|s_1)$ is the likelihood of s_1 being transmitted and $p(z|s_2)$ is the likelihood of s_2 being transmitted.



The detector would then take the decision based on the following criteria

If $z(t) > \gamma$ then assume H_1

If $z(t) < \gamma$ then assume H_2

Here H_1 is hypothesis 1 say meaning signal 1 i.e. s_1 is transmitted & hence received bit is 1.
 H_2 is hypothesis 2 say meaning signal 2 i.e. s_2 is transmitted and hence received bit is 0.

Here γ is the threshold level for the binary decisions. ~~base~~ Selection of this γ is popularly done with the idea of minimizing the error. Let's say, the minimum error value is $\gamma = \gamma_0$. A small analysis based on Baye's theorem of probability may be formulated as

$$\frac{P(z|s_1)}{P(z|s_2)} \underset{H_2}{\overset{H_1}{\geq}} \frac{P(s_2)}{P(s_1)}$$

where $P(s_2)$ and $P(s_1)$ are previously known probabilities of transmission of s_2 & s_1 respectively. This is also called "a priori" probability.

For minimum error, we assume $P(s_1) = P(s_2)$ and $p(z|s_1)$ and $p(z|s_2)$ are symmetrical,

$$z(T) \underset{H_2}{\overset{H_1}{\geq}} \frac{a_1 + a_2}{2} = \gamma_0$$

i.e. the threshold level for minimum error is $\gamma_0 = \frac{a_1 + a_2}{2}$. This is called optimum threshold.

Another way to look at it is that when $P(s_1) = P(s_2)$ i.e. equally probable signals are transmitted, the decision simply becomes based on

$$P(z|s_1) \underset{H_1}{\overset{H_2}{\geq}} P(z|s_2)$$

The receiver that minimizes the error is called ~~max~~ maximum likelihood receiver.

Error Probability

(41)

There are two possibilities of occurrence of error.

- ① s_1 was sent and $z(T) < \gamma_0$ i.e. 1 was sent and 0 was received.
- ② s_2 was sent and $z(T) > \gamma_0$ i.e. 0 was sent and 1 was received

Probability of such occurrence is

$$P(e|s_1) = P(H_2|s_1) = \int_{-\infty}^{\gamma_0} p(z|s_1) dz \text{ and}$$

$$P(e|s_2) = P(H_1|s_2) = \int_{\gamma_0}^{\infty} p(z|s_2) dz$$

For total probability, we take the sum of all cases. Again by Baye's theorem

$$P_B = \sum_{i=1}^2 P(e|s_i) P(s_i)$$

$$= P(e|s_1) P(s_1) + P(e|s_2) P(s_2)$$

$$\text{or } P_B = P(H_2|s_1) P(s_1) + P(H_1|s_2) P(s_2)$$

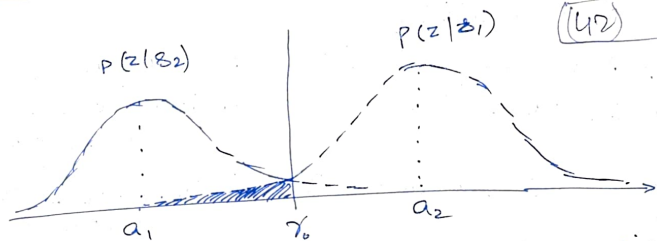
Again for equiprobable signals

$$P(s_1) = P(s_2) = \frac{1}{2}$$

$$\Rightarrow P_B = \frac{1}{2} P(H_2|s_1) + \frac{1}{2} P(H_1|s_2)$$

Also, as the two pdfs are symmetrical,

$$P_B = P(H_2|s_1) = P(H_1|s_2)$$



Thus the probability of a bit ~~is~~ error, P_B is numerically equal to the area of the tail of either likelihood function $p(z|s_1)$ or $p(z|s_2)$ falling on the incorrect side of the threshold. Thus to compute this, we may use either function.

$$P_B = \int_{\gamma_0 = \frac{a_1 + a_2}{2}}^{\infty} p(z|s_2) dz.$$

~~Using~~ Using the formula we get

$$P_B = \int_{\gamma_0}^{\infty} \frac{1}{\sigma_0 \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{z - a_2}{\sigma_0} \right)^2} dz$$

where σ_0^2 is variance of noise.

Assuming $u = \frac{z - a_2}{\sigma_0}$

$$P_B = \int_{u = \frac{a_1 - a_2}{2\sigma_0}}^{u = \infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right)$$

Here $Q(x)$ is called complementary error function.

$Q(x)$ may be defined as

$$Q(x) = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$$

For a value of $x > 3$, $Q(x)$ can be approximated as

$$Q(x) \approx \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

If in case you are ~~are~~ provided with a table ^{or values} for $\text{erfc}(x)$ function, you can use the following relation

$$Q(x) = \frac{1}{2} \text{erfc}\left(\frac{x}{\sqrt{2}}\right)$$

and ~~see~~ $\text{erfc}(x) = 2Q(x\sqrt{2})$

i.e. say you want to find

$Q(x)$ for $x = \sqrt{2}$

then $Q(\sqrt{2}) = \frac{1}{2} \text{erfc}\left(\frac{\sqrt{2}}{\sqrt{2}}\right)$

$$= \frac{1}{2} \text{erfc}(1)$$

$$= \frac{1}{2} (0.1573)$$

$$Q(\sqrt{2}) = 0.079$$

Even if $\text{erf}(u)$ is available
 $\text{erfc}(u) = 1 - \text{erf}(u)$

Matched Filter

(45)

It is a linear filter designed to provide maximum signal-to-noise power ratio at its output for a given transmitted symbol waveform.

Let us consider a known signal $s(t)$ plus AWGN $n(t)$ is input to an LTI filter followed by a sampler. At time $t = T$, the sampler output consists of a signal component a_i and a noise component n_o . Variance of output noise is σ_o^2 so that instantaneous (SNR)_T at time $t = T$ will be

$$\left(\frac{S}{N}\right)_T = \frac{a_i^2}{\sigma_o^2}$$

What we need is a filter transfer function $H(f)$ that maximises this equality.

$a_i(t)$ can be represented as

$$a_i(t) = \int_{-\infty}^{\infty} H(f) S(f) e^{j2\pi f t} df$$

Also, assuming that the noise has a PSD of $\frac{N_o W_{Hz}}{2}$

then
$$\sigma_o^2 = \frac{N_o}{2} \int_{-\infty}^{\infty} |H(f)|^2 df$$

$$\Rightarrow \left(\frac{S}{N}\right)_T = \frac{\left| \int_{-\infty}^{\infty} H(f) S(f) e^{j2\pi f T} df \right|^2}{\frac{N_o}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

Here we go to a mathematical tool called the Schwarz's Inequality. (UG)

It states that if $f_1(x) = K f_2^*(x)$

then

$$\left| \int_{-\infty}^{\infty} f_1(x) f_2(x) dx \right|^2 \leq \int_{-\infty}^{\infty} |f_1(x)|^2 dx \int_{-\infty}^{\infty} |f_2(x)|^2 dx$$

Here if we take $f_1(x) = H(t)$
and $f_2(x) = S(t) e^{j2\pi f t}$

$$\begin{aligned} \text{then } \left| \int_{-\infty}^{\infty} H(t) S(t) e^{j2\pi f t} dt \right| &\leq \int_{-\infty}^{\infty} |H(t)|^2 dt \int_{-\infty}^{\infty} |S(t) e^{j2\pi f t}|^2 dt \\ &\leq \int_{-\infty}^{\infty} |H(t)|^2 dt \int_{-\infty}^{\infty} |S(t)|^2 dt \end{aligned}$$

$$\Rightarrow \left(\frac{S}{N} \right)_T \leq \frac{2}{N_0} \int_{-\infty}^{\infty} |S(t)|^2 dt$$

or $\max \left(\frac{S}{N} \right)_T = \frac{2E}{N_0}$

where $E = \int_{-\infty}^{\infty} |S(t)|^2 dt$

But this holds only when

$$H(t) = H_0(t) = K S^*(t) e^{-j2\pi f T}$$

$$\text{or } h(t) = F^{-1} \{ K S^*(f) e^{-j2\pi f T} \} \quad (47)$$

$\Rightarrow$ Since $S(t)$ is a real value signal, we can write

$$h(t) = \begin{cases} K S(T-t) & 0 \leq t \leq T \\ 0 & \text{elsewhere.} \end{cases}$$

Hence, the impulse response of a filter that produces the maximum output SNR is the mirror image of the message signal $S(t)$ delayed by the symbol time duration T .

*Note: This time delay actually makes the response causal)

Response of a matched filter

The response $z(t)$ can be seen as the convolution of $z(t)$ and $h(t)$

$$\Rightarrow z(t) = \int_{-\infty}^{\infty} z(\tau) h(t-\tau) d\tau$$

Using the equation for $h(t)$ for optimum response

$$\begin{aligned} z(t) &= \int_{-\infty}^{\infty} z(\tau) K S(T-(t-\tau)) d\tau \\ &= \int_{-\infty}^{\infty} z(\tau) S(T-t+\tau) d\tau \end{aligned}$$

$\therefore$ At sample time $t=T$
 $z(T) = \int_{-\infty}^{\infty} z(\tau) S(\tau) d\tau$

which can be seen as the correlation of $z(t)$ & $S(t)$.

Thus, any $s_i(t)$ whose correlation with $r(t)$ gives the maximum $z_i(t)$ is the signal that matched $z_2(t)$ better than all the other $s_j(t)$ (where $i \neq j$)

Optimum Receiver

Optimum Receiver is one in which we select the optimum (matched) receiving filter and the optimum (maximum likelihood) decision threshold.

By the criteria used in maximum likelihood case, we observed that

$$P_B = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right)$$

To minimize the error we need a system that maximizes $\frac{a_1 - a_2}{2\sigma_0}$ or in terms of SNR, that maximizes

$$(SNR)_T = \frac{(a_1 - a_2)^2}{\sigma_0^2}$$

$$\sigma_0^2 = N_0/2$$

$$(a_1 - a_2)^2 = E_d \quad (\text{Energy of the difference signal})$$

$$E_d = \int_0^T [s_1(t) - s_2(t)]^2 dt$$

Assuming that the ~~input~~ filter is matched to the input difference signal,

$$(SNR)_T = \frac{(a_1 - a_2)^2}{\sigma_0^2} = \frac{2E_d}{N_0} \quad (149)$$

* Note: Maximizing the signal difference maximizes the distance between the two inputs.

$$\Rightarrow P_B = Q\left(\frac{\sqrt{2E_d}}{\sqrt{2N_0}}\right) = Q\left(\sqrt{\frac{E_d}{2N_0}}\right)$$

Here evaluating E_d

$$\begin{aligned} E_d &= \int_0^T [s_1(t) - s_2(t)]^2 dt \\ &= \int_0^T s_1^2(t) dt + \int_0^T s_2^2(t) dt - 2 \int_0^T s_1(t) s_2(t) dt \end{aligned}$$

For maximum likelihood we assumed both bits have same energy +

$$\Rightarrow E_B = \int_0^T s_1^2(t) dt = \int_0^T s_2^2(t) dt$$

Also, we define a variable γ , which is the cross-correlation coefficient, such that

$$\Rightarrow \gamma = \frac{1}{E_B} \int_0^T s_1(t) s_2(t) dt$$

* Note - In case the signals are defined in vector space, γ may be defined as the cosine of the angle between them.
i.e. if s_1 & s_2 have angle θ b/w them,

$$\gamma = \cos \theta \Rightarrow -1 \leq \gamma \leq 1$$

$$\text{Now } E_d = 2E_b - 2E_b\gamma$$

$$= 2E_b(1 - \gamma)$$

Using this we may say

$$P_B = Q\left(\sqrt{\frac{E_b(1-\gamma)}{N_0}}\right)$$

perfectly correlated or

Case 1: If we have $\gamma = 1$ i.e. matched signals.

$P_B = Q(0) = 0.5$
i.e. 50% probability that there will be a bit error. This makes it unusable.

Case 2: Antipodal or Anticorrelated signals
i.e. Angle is 180° .
 $\Rightarrow \gamma = -1$ $P_B = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

Case 3: Orthogonal signals i.e. angle is 90°
 $\Rightarrow \gamma = 0$ (uncorrelated)

$$P_B = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

In vector space, if each bit has energy E_b and we assume signal amplitude is $\sqrt{E_b}$, we may plot

$$\frac{|s_1| = \sqrt{E_b}}{|s_2| = \sqrt{E_b}} \quad \text{correlated.}$$

$$\sqrt{E_b} = |s_1| \quad |s_2| = \sqrt{E_b} \quad \text{Antipodal}$$

$$|s_2| = \sqrt{E_b} \quad \sqrt{2E_b} \quad |s_1| = \sqrt{E_b} \quad \text{orthogonal}$$

Hence for 2 signals only, we should use antipodal signals for the best result.

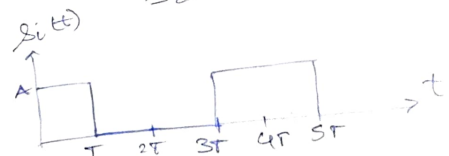
Error probability performance of binary

Signals

(Unipolar)

$$s_1(t) = A \quad 0 \leq t \leq T$$

$$s_2(t) = 0 \quad 0 \leq t \leq T$$



This is clearly an orthogonal set as

$$\varphi = 0$$

$$|s_1(t) - s_2(t)| = A$$

Assuming that $s_1(t)$ was transmitted,

$$a_1(T) = E\{z(T) | s_1(t) \text{ transmitted}\}$$

(Expected value of $z(T)$ if $s_1(t)$ is transmitted)

$$= E\left\{\int_0^T (A^2 + A n(t)) dt\right\}$$

$$= A^2 T \quad (\text{as Expected noise is zero})$$

Similarly $a_2(t) = E\{z(T) | s_2(t) \text{ transmitted}\}$

$$= 0$$

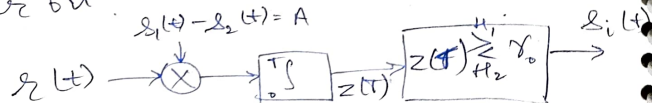
$$\Rightarrow \gamma_0 = \left(\frac{a_1 + a_2}{2}\right) = \frac{A^2 T}{2}$$

And $P_B = Q\left(\sqrt{\frac{E_b}{2N_0}}\right) = Q\left(\sqrt{\frac{A^2 T}{2N_0}}\right)$

$$= Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

where $E_b = \frac{A^2 T}{2}$ is the average energy per bit

Per bit



~~Int~~

② Bipolar Signals

In this case

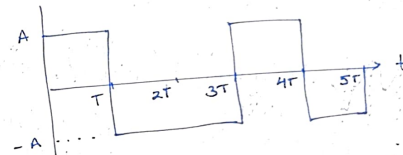
$$s_1(t) = +A$$

$0 \leq t \leq T$ for Bih-1

$$s_2(t) = -A$$

$0 \leq t \leq T$ for Bih-0

$$\Rightarrow s_1(t) = -s_2(t) \quad \text{i.e. Antipodal signals.}$$



The received bit levels will be a_1 & a_2 and thus $a_1 = -a_2$. Here, the threshold level ~~is~~ ^{is} ~~zero~~ ^{is} thus $\gamma_0 = \frac{a_1 + a_2}{2} = 0$.

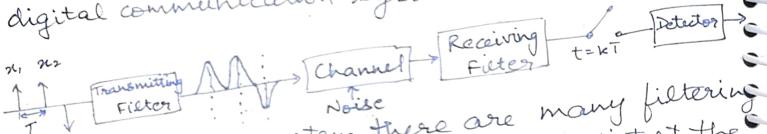
$z(T) (= z_1(T) - z_2(T))$, if positive, then the signal received is ~~$s_1(t)$~~ $s_1(T)$ and if -ve, then the signal received is ~~$s_2(t)$~~ $s_2(T)$.

Again $P_B = Q\left(\sqrt{\frac{E_d}{2N_0}}\right)$, $E_d = (2A)^2 T$ & $E_b = A^2 T$

$$\Rightarrow P_B = Q\left(\sqrt{\frac{2A^2 T}{N_0}}\right) \Rightarrow P_B = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

Inter Symbol Interference

Lets talk about the filtering aspects of a typical digital communication system.



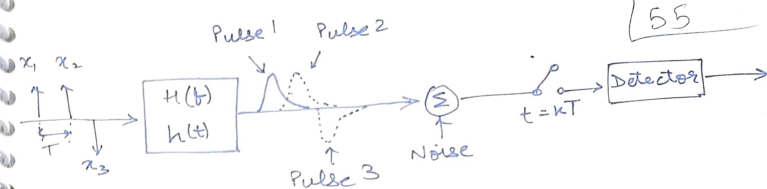
Throughout the system there are many filtering (and reactive circuit elements) that distort the original signal.

Even when we assume absence of noise, the filtering characteristics still are capable enough to cause significant distortion.

Assuming transmitter is modelled using a transfer function $H_t(f)$, the channel is modelled using a transfer function $H_c(f)$ and the receiver is modelled using the transfer function $H_r(f)$, then the overall equivalent system transfer function is

$$H(f) = H_t(f) H_c(f) H_r(f)$$

and a simplified representation can be



As observed in the diagram above, the pulse which are received can overlap each other over the range of transmission as a result of broadening due to filtering process.

The tail of such a pulse can "smear" into the adjacent symbol pulse intervals thereby causing interference in the detection process and degrading the system performance. Such interference is called Inter Symbol Interference (ISI).

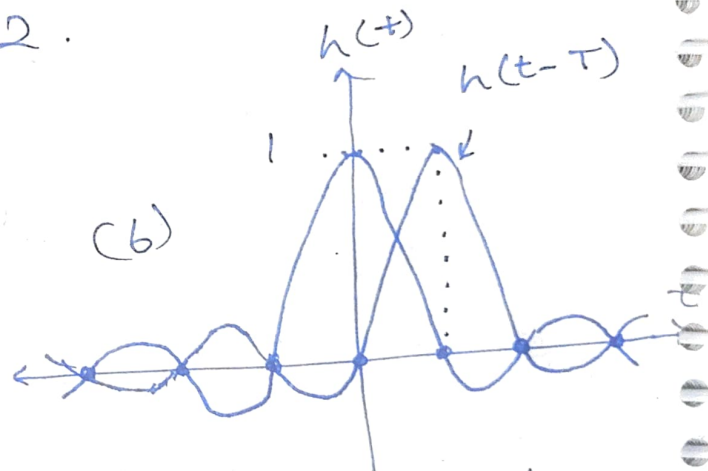
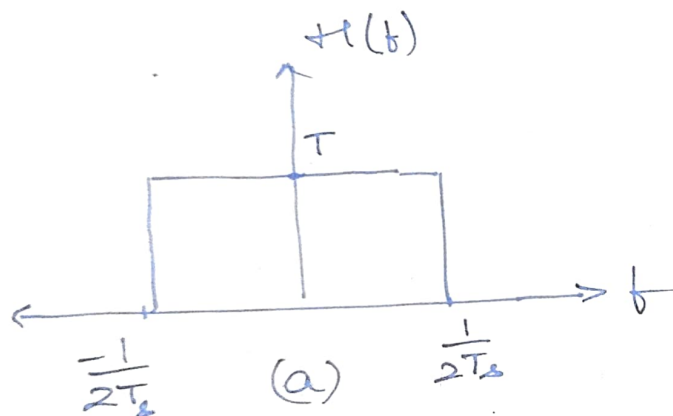
The process of controlling the ISI thus remains with the criteria of controlling $H(f)$.

In case $H_t(f)$ and $H_c(f)$ are specified, $H_r(f)$ is then modelled to remove the error due to both these systems. Such a receiving filter is called an "Equalizing Filter" or a "Receiving Equalizing Filter" in case both the actions are done using the same filter.

Another way is to model $H_t(f)$ and $H_r(f)$ for a specified $H_c(f)$.

Nyquist Zero ISI Model

Nyquist showed that if $h(t)$ is made into a rectangular pulse with ~~sig~~ single sideband & single sided bandwidth of $\frac{1}{2T_s}$ i.e. total duration of $\frac{1}{T_s}$ and amplitude T_s , i.e. an ideal Nyquist filter, then it is possible to generate and detect digital waveforms with zero ISI. In other words, theoretical minimum bandwidth required to transmit R_s symbols/sec without ISI is $R_s/2$.

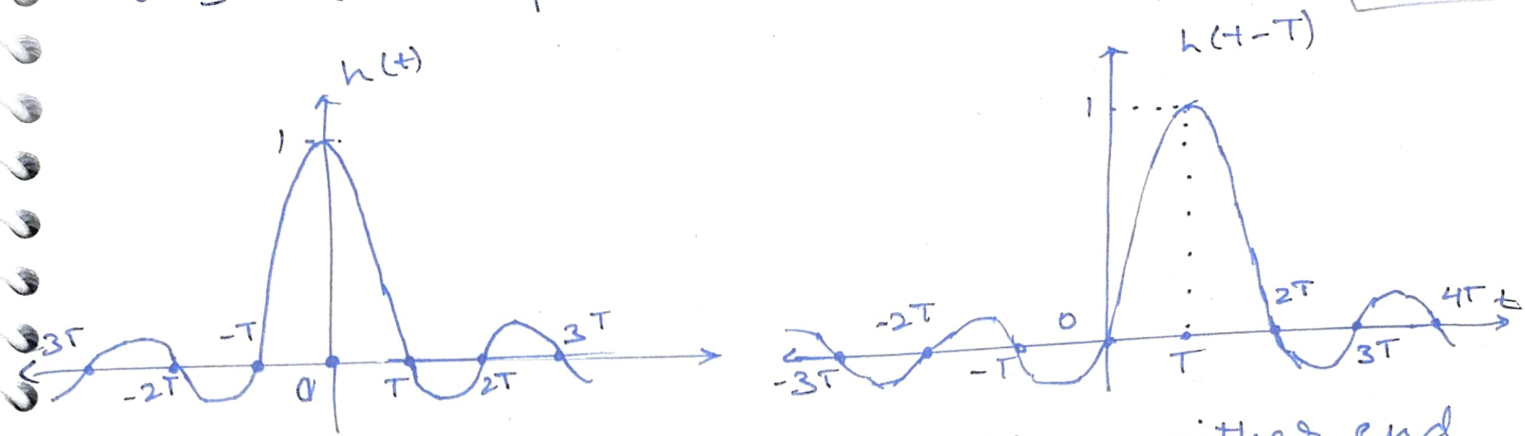


In time domain, the system $h(t)$ would then be of the form $h(t) = \text{sinc}\left(\frac{t}{T}\right)$

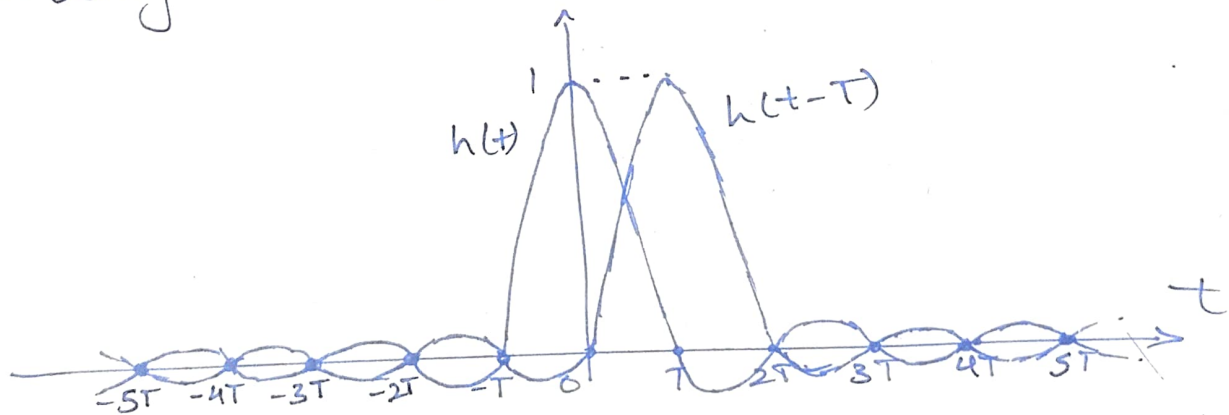
This pulse shape of $\text{sinc}\left(\frac{t}{T}\right)$ is called an ideal Nyquist Pulse.

Nyquist established that if each pulse is of the form $\text{sinc}\left(\frac{t}{T}\right)$, then the pulses can be detected without ISI. The same can be observed by examining the figure (b) above.

Consider two pulses $h(t)$ and $h(t-T)$. 57



Here, even though $h(t)$ has tails on either end running for ∞ length, at ~~every~~ ^{the} instant $t=T$, when the pulse $h(t-T)$ is to be sampled, value of $h(t)$ is equal to zero. In fact the value of any arbitrary ^{symbol} pulse $h(t-kT)$ remains zero at all sample points except $t=kT$. Hence there is no chance of any symbol interfering in any other symbol's detection.



For baseband systems: The bandwidth required to detect $\frac{1}{T}$ such symbols (pulses) is equal to $\frac{1}{2T}$. In other words a system with bandwidth $W = \frac{1}{2T} = \frac{R_s}{2}$ can support a maximum transmission rate of $2W = \frac{1}{T} = R_s$ Symbols/sec without ISI. This is called the Nyquist Bandwidth constraint.

Important definitions in comm systems:

① Symbol rate packing = $\frac{R}{W}$

② Bandwidth Efficiency = $\frac{R \times K}{W}$

where R = Symbol rate or Band Rate

W = Bandwidth.

K = No. of bits / Symbol.

Note: Symbol rate packing is evaluated in Symbols/s/Hz
Bandwidth Efficiency is evaluated in bits/s/Hz

As is evident from the rectangular shaped transfer function of the ideal ~~transfer~~ Nyquist filter and the infinite length of its corresponding pulse, that such ideal filters are not realizable; they can only be approximated.

Although there can be a complete family of "Nyquist filters" ^{and} "Nyquist Pulses" which satisfy the zero ISI criteria, the most popular ~~one~~ ^{are} filters are raised cosine and root-raised cosine filters.

Note: Investigating the symbol rate-bandwidth constraints

As it has been proven that ideal $\text{sinc}(\frac{t}{T})$ form leads to zero ISI, mathematically:

$$h(t) = \text{sinc}\left(\frac{t}{T}\right) = \text{sinc}(Rt) \quad \left(\frac{1}{T} = R \text{ symbol rate}\right)$$

$$\Rightarrow F[h(t)] = H(f) = \frac{1}{R} \text{rect}\left(\frac{f}{R}\right) \\ = T \text{rect}\left(\frac{f}{R}\right) \\ = T \text{rect}(Tf)$$

Hence ~~the~~ two sided bandwidth equals $\frac{1}{T}$
and one sided bandwidth equals $\frac{1}{2T}$.

Raised Cosine Filter

59

One frequently used Nyquist class filter with transfer function:

$$H(f) = \begin{cases} 1 & \text{for } |f| < 2W_0 - W \\ \cos^2\left(\frac{\pi}{4} \left| \frac{|f| + W - 2W_0}{W - W_0} \right| \right) & \text{for } 2W_0 - W < |f| < W \\ 0 & \text{for } |f| > W \end{cases}$$

where $W_0 = \frac{1}{2T}$ is the minimum required Nyquist Bandwidth

W = Absolute bandwidth

W_0 is also the -6dB bandwidth (half amplitude point) of the raised cosine spectrum

$W - W_0$ is called the Excess Bandwidth.
i.e. additional bandwidth beyond nyquist min.

We also define $r = \frac{W - W_0}{W_0}$ as the roll-off factor.
(Also called fractional excess BW)

$$\Rightarrow 0 \leq r \leq 1$$

$r = 0$ represents ideal case where $W = W_0$
 $r = 1$ represents the case where $W = 2W_0$
i.e. usage of double the bandwidth for transmission.

* Note:

Some books define raised cosine function as

$$H(f) = \begin{cases} T & \text{for } 0 \leq |f| \leq \frac{1-B}{2T} \\ \frac{T}{2} \left\{ 1 + \cos \left[\frac{\pi T}{B} \left(|f| - \frac{1-B}{2T} \right) \right] \right\} & \text{for } \frac{1-B}{2T} \leq |f| \leq \frac{1+B}{2T} \\ 0 & \text{for } |f| > \frac{1+B}{2T} \end{cases}$$

where T = Symbol time
 B = roll off factor.

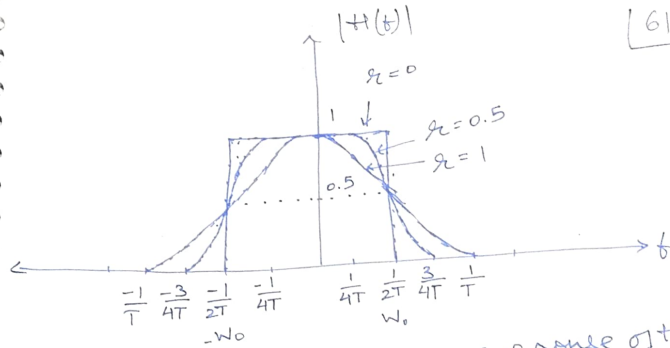
Here I have observed an error in their definition by the factor of multiplication by T . Otherwise putting $T = \frac{1}{2W_0}$ and $B = \frac{W - W_0}{W_0}$, we can get the

same expression described earlier. The impulse response of this filter is correspondingly

$$h(t) = \frac{\sin\left(\frac{\pi t}{T}\right) \cos\left(\frac{\pi t B}{T}\right)}{\pi t \left(1 - \frac{4B^2 t^2}{T^2}\right)}$$

Again by substituting $B = \frac{W - W_0}{W_0}$, $T = \frac{1}{2W_0}$ and the formula $\frac{\sin(\pi x)}{\pi x} = \text{sinc}(x)$ we can get the

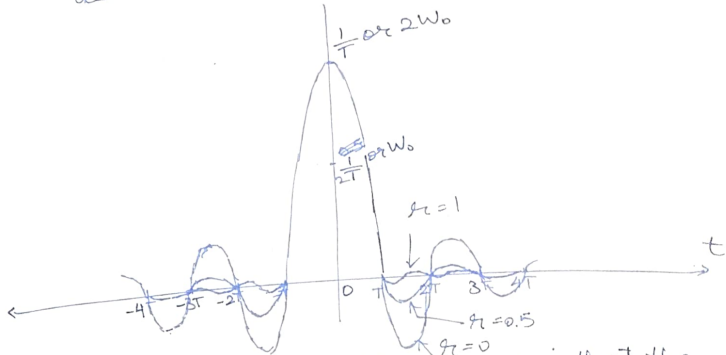
same impulse response as for the filter defined earlier. There is no error observed in this



The corresponding impulse response of this transfer function is

$$h(t) = 2W_0 \text{sinc}(2W_0 t) \left(\frac{\cos[2\pi(W - W_0)t]}{1 - [4(W - W_0)t]^2} \right)$$

The plot of the system response can be drawn as



What can be noted in all cases of η is that the value of the pulse remains same at each sample time thus satisfying the Nyquist criteria.

For the same reason as ideal Nyquist pulses, raised cosine spectrum is not physically realizable. A filter must have an impulse response of finite duration and exhibit a zero output prior to the pulse turn on time which is not the case with the family of raised cosine characteristics. 62

* These unrealisable filters are NONCAUSAL since the filtered pulse begins at $t = -\infty$ and is of infinite duration.

The absolute bandwidth in term of roll off can be written as

$$W = (1+\alpha)W_0$$

$$\text{or } W = \frac{1}{2}(1+\alpha)R_s$$

It must be noted here that for $R_s > 0$, there is bandwidth expansion beyond the Nyquist minimum. Thus for that case R_s is less than twice the bandwidth. The demodulator is giving one sample per symbol, and hence, the Nyquist sampling theorem is violated. Hence we are left with too few samples to reconstruct the analog waveform unambiguously.

But in digital communication, it is not our aim to reconstruct the analog wave. Since we are able to achieve zero ISI at the time when symbol is sampled, we can still have unambiguous detection.

Bandpass modulated systems such as ASK, PSK require double the bandwidth in comparison to the equivalent baseband signals. These signals are often called Double Sideband (DSB) signals. Thus ASK and PSK signals have the required DSB bandwidth W_{DSB} given by 63

$$W_{DSB} = (1+\alpha)R_s.$$

Recall here that the receiver filter is compensating the effects of the transmitter and channel filtering processes. If we assume that channel induced ISI is negligible, then this compensation can be achieved by choosing the transmitter and receiver filters to be matched in nature and equal to the square root of the raised cosine filter. This is then known as the root raised cosine. The product of these root raised cosine filters result in composite raised cosine system transfer function.

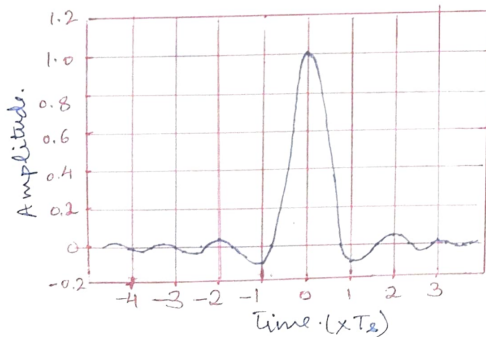
* Sometimes, to reduce the channel induced noise, equalizing filters are also used at the receiver. In this case the equalizing and receiving filters must be designed to compensate the effects of both the transmitter and channel induced ISI, so as to yield an overall system transfer function characterized by zero ISI.

* Trade off between Roll off and Bandwidth.

If we assume high roll off factor, i.e. $\alpha \rightarrow 1$, then we observe that the time domain pulse has smaller tails which exhibits less sensitivity to timing errors and small degradation due to ISI. But the cost is the excess bandwidth it acquires. [64]

On the other hand smaller roll off means smaller the excess bandwidth thereby allowing more number of users that can simultaneously use the system. The cost is longer pulse tails ~~and the~~ with larger amplitudes thus greater sensitivity to timing errors.

The impulse response of a root raised cosine filter called the "Square root Nyquist pulse" is shown below. The peak value is normalized to unity with a filter roll off $\alpha = 0.5$.



In comparison to the actual Nyquist pulse, the Square root Nyquist pulse, although similar in shape, makes slightly faster transitions, thus its spectrum does not decay as rapidly as the spectrum of the raised cosine filter i.e. Nyquist pulse. Another important difference is that the Square root Nyquist pulse DOES NOT exhibit zero ISI. It can be verified from the figure that at sample times, the impulse response does not have zero amplitude. However, if a root raised cosine filter is used at both the transmitter and the receiver, the product of these transfer functions being a raised cosine, will give rise to an output having zero ISI. [65]

Demodulation / Detection of Shaped Pulses.

The demodulation process of the shaped pulse firstly involves the matched filter system followed in root raised cosine systems. But a brief review may be made regarding the difference in the roles of matched filters and conventional filters.

Matched vs Conventional filters

The role of conventional filters is to screen out unwanted spectral components of a received signal while maintaining some measure of quality for signals occupying a ~~selected~~ selected span of the spectrum, called the pass band. In the pass band the system is assumed to be rather designed to have a uniform gain and a linear phase to frequency characteristics. Over the remaining spectrum, called the stop band, these filters have some minimum attenuation.

The priority of a matched filter is totally different. The priority is to maximize the SNR of a known signal in the presence of AWGN. So the ~~first~~ difference is

I Conventional filters are applied to random signals defined only by their bandwidth while matched filters are applied to known signals with random parameters (such as amplitude).

Considering the fact that matched filter is like a template matched to the known shape of the signal being processed, another important difference can be taken out.

II

A conventional filter tries to protect the spectral structure of the signal of interest. On the other hand a matched filter significantly changes the spectral structure by gathering the signal energy matched to its template, and at the end of each symbol time, presenting the result as a peak amplitude.

67

In general, a digital communication receiver uses both kinds of filters to process the received signals. In this case, the work of a conventional filter is to produce a high quality estimate of the signal for presentation to the filter. The matched filter gathers the received signal energy, and when its output is sampled at $(t=T)$, a voltage proportional to that energy is produced for subsequent detection processing.

Channel Characterization

Many communication channels (eg. Telephone & wireless) can be characterized as band-limited linear filters with an impulse response $h_c(t)$ and a frequency response

$$H_c(f) = |H_c(f)| e^{j\phi_c(f)}$$

where $h_c(t)$ and $H_c(f)$ are FT pairs, $|H_c(f)|$ is channels amplitude response and $\phi_c(f)$ is channels phase response. In order to achieve ideal transmission characteristics over a channel,

the amplitude response $|H_c(f)|$ must remain constant over the signal bandwidth W . Also $\angle H_c(f)$ must be a linear function of frequency. If $|H_c(f)|$ is not constant, there is amplitude distortion and if $\angle H_c(f)$ is not linear, there is phase distortion. In many channels both these distortions occur simultaneously causing frequency spreading and leading to ISI. Thus a signal processing or filtering technique designed to reduce or eliminate ISI is required. Such operations are called equalization techniques. [68]

Equalization techniques may be divided into two main categories.

1. Maximum Likelihood Sequence Estimation

This entails making measurements of $h_c(t)$ and then providing a means for adjusting the receiver to the transmission environment.

The goal here is not to reshape or directly compensate the distorted samples in any way. Instead the technique for MLSE ~~is~~ receiver is to adjust ~~the~~ itself in such a way that it can better deal with the distorted samples. eg. ~~Viterbi~~ Viterbi Equalization for fading channels.

2. Equalization with filters [69]

This technique uses ~~equalizers~~ ^{filters} to compensate the distorted pulses, modifying them to remove the effects of ISI. This technique is more popularly used. It can further be classified as

- (a) Feed forward element filters or transversal equalizers which contain feedforward elements only. (Linear devices)
- (b) Decision Feedback equalizers which contain both feedforward and feedback elements. (Nonlinear devices).

Further classification of DFEs can be done as:

(1) Preset or Adaptive: based on nature of operation

(2) Symbol spaced or Fractionally spaced:

Symbol spaced $\rightarrow$ Samples exist only at symbol boundaries

Fractionally spaced $\rightarrow$ Multiple samples exist in a ~~single~~ single symbol time.

Furthering our discussion on the performance of a ~~root~~ Raised Cosine filter in removing ISI it is found that if the complete transfer function of the transmission system is the Raised Cosine function, then the net system can be given by

$$H_{rc}(f) = H_t(f) H_c(f) H_e(f) H_r(f)$$

Here $H_{RC}(t) =$ Raised cosine filter TF

$H_t(t) =$ Transmitter TF

$H_c(t) =$ Channel TF

$H_e(t) =$ Equalizer TF

$H_r(t) =$ Receiver TF.

We also know that if $H_t(t)$ and $H_r(t)$ are matched filters to root raised cosine filter, then

$$H_{RC}(t) = H_t(t) H_r(t)$$

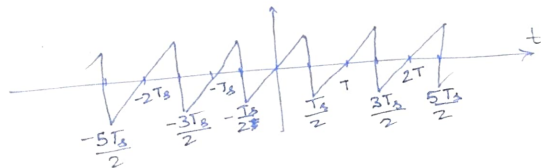
$\therefore$ The relation between Equalizer and channel characteristics must be

$$H_e(t) H_c(t) = 1$$

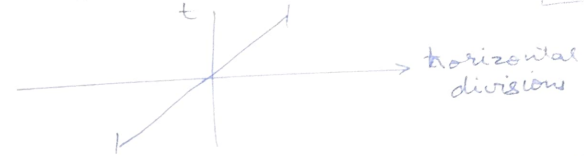
or $H_e(t) = \frac{1}{|H_c(t)|} e^{-j\phi_c(t)}$

Eye Pattern

It is a technique of representing the baseband signals in a specific way. On the vertical plates of an oscilloscope we connect the receiver's response to any random sequence. On the horizontal plates we connect a sawtooth wave at the signalling frequency i.e. a sawtooth with period equal to symbol time or duration.



In a conventional CRO, the horizontal plates are fed with a unit ramp system limited at particular values set by horizontal range settings.



When we use a sawtooth instead, the scope range is set to symbol time T_s . Hence repeated plots over the symbol time are ~~data~~ observed on the CRO which appear superimposed onto each other. Consider the following figures.

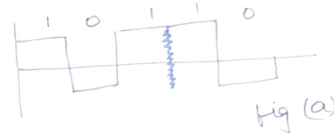


fig (a)

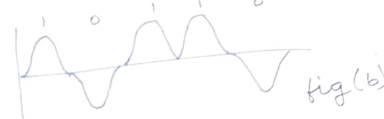
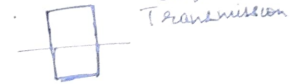


fig (b)

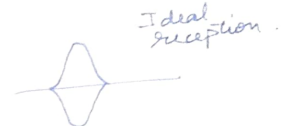
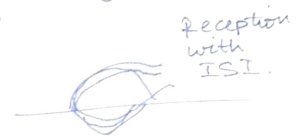
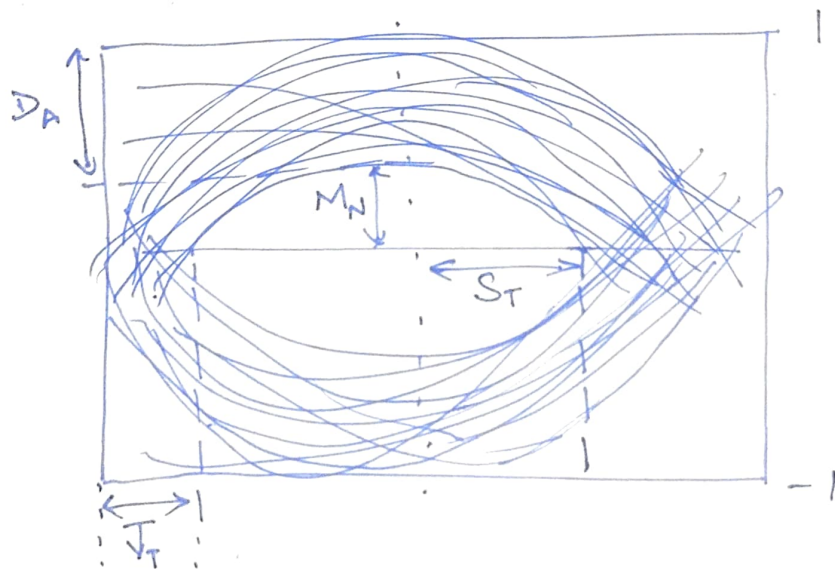


fig (c)



This pattern when observed looks like an eye.
Thus known as eye pattern. [72]

Therefore "bigger the eye opening", less is the error due to ISI and vice versa.



A typical Eye diagram is shown above which indicates

- ① D_A : Range of amplitude differences as a measure of distortion caused by the ISI.
 - ② J_T : Range of time differences of the zero crossings as a measure of timing jitter.
 - ③ M_N : Noise margin
 - ④ S_T : Sensitivity to timing error.
-